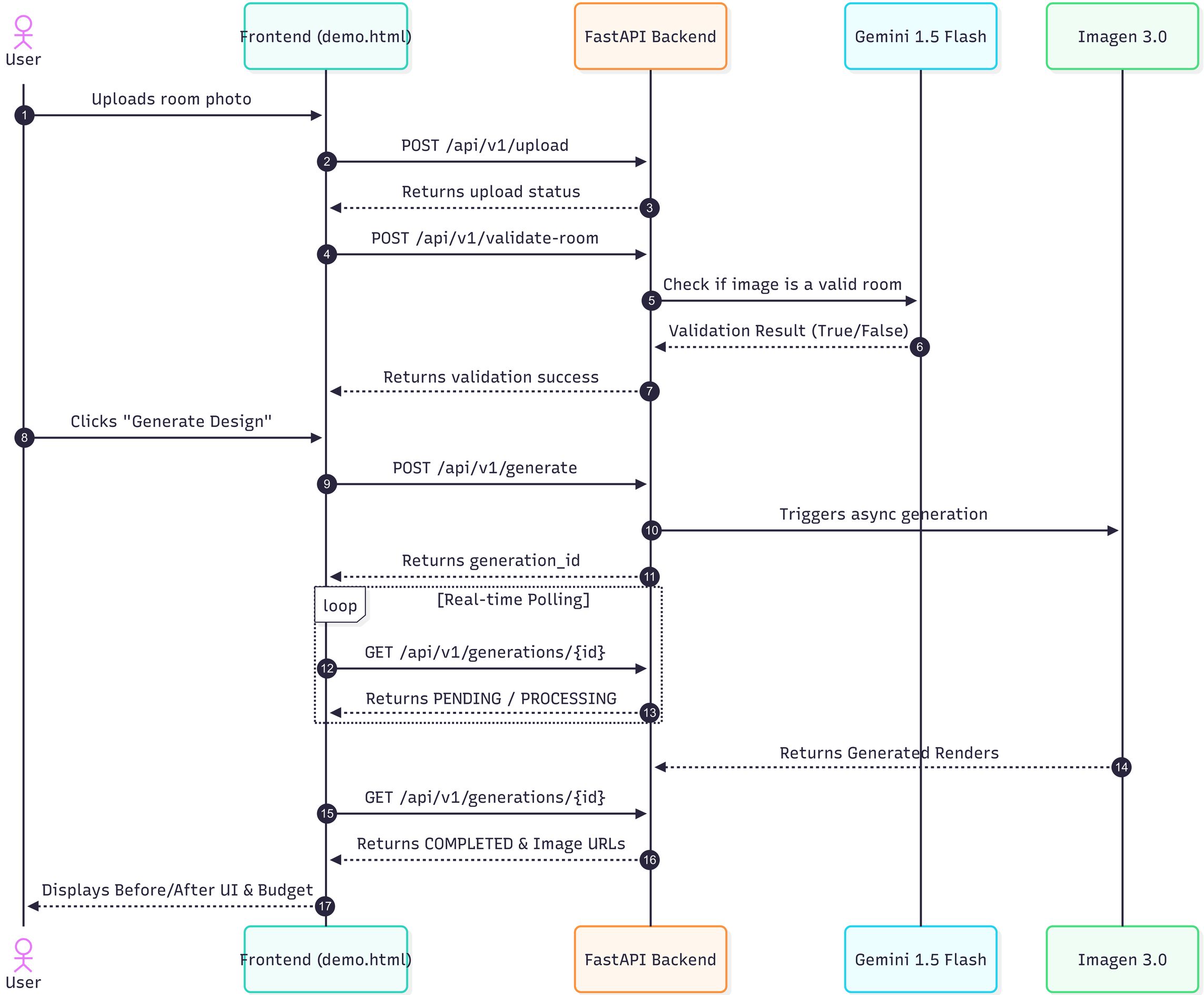
User
Frontend (demo.html)
FastAPI Backend
Gemini 1.5 Flash
Imagen 3.0
1 Uploads room photo
2 POST /api/v1/upload
3 Returns upload status
4 POST /api/v1/validate-room
5 Check if image is a valid room
6 Validation Result (True/False)
7 Returns validation success
8 Clicks "Generate Design"
9 POST /api/v1/generate
10 Triggers async generation
11 Returns generation_id
loop
[Real-time Polling]
12 GET /api/v1/generations/{id}
13 Returns PENDING / PROCESSING
14 Returns Generated Renders
15 GET /api/v1/generations/{id}
16 Returns COMPLETED & Image URLs
17 Displays Before/After UI & Budget
User
Frontend (demo.html)
FastAPI Backend
Gemini 1.5 Flash
Imagen 3.0